

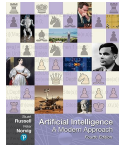


MSML610: Advanced Machine Learning

Lesson 03.1: A Brief History of AI

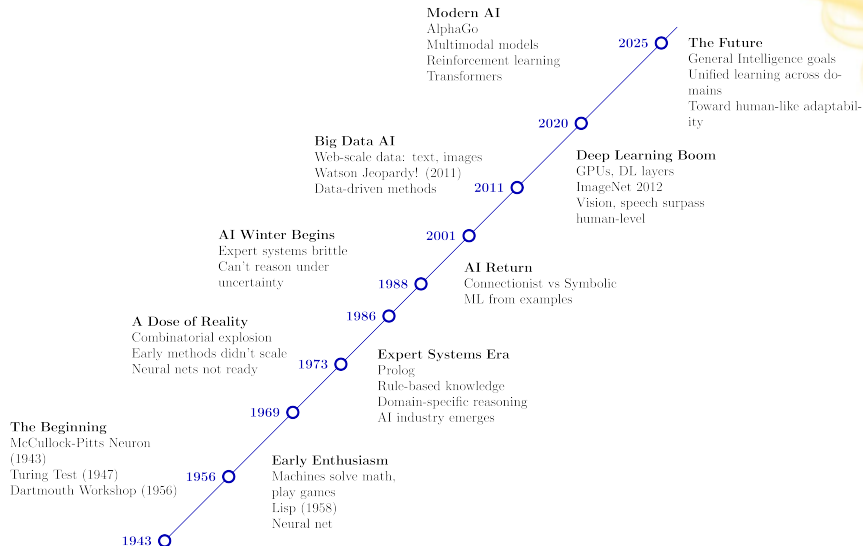
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References: - AIMA (Artificial Intelligence: a Modern Approach), Chap 1



- ***Brief History of AI***
- Risks and Benefits of AI

AI Timeline



The Beginning (1943-1956)

- **Artificial neuron**
 - Model (McCulloch-Pitts 1943) based on:
 - Brain physiology
 - Propositional logic
 - Compute any function with connected neurons
 - Neuron on/off based on stimulation from neighboring neurons
 - Implement logical AND, OR, NOT with simple neuron networks
- **Alan Turing, 1947**
 - Turing test, machine learning, reinforcement learning
 - Create human-level AI
 - Develop learning algorithms
 - Teach machine like a child
- **Birth of AI**
 - McCarthy organized first AI workshop (1956)
 - The Logic Theorist (Newell and Simon, 1956)
 - Programs to “think non-numerically” and prove theorems

Enthusiasm and Great Expectations (1952-1969)

- **Early years of AI were full of successes**
 - Before computers could only do arithmetics
 - “A machine can never do X (e.g., games, puzzles, IQ tests, ...)”
 - AI researchers showed machines could do one X after another
- **General Problem Solver**
 - Imitate human problem-solving
 - Consider sub-goals and possible actions
- **Program learned to play checkers**
 - Use reinforcement learning from victories and mistakes
- **Lisp (1958)**
 - High-level language used for 30 years in AI
- **First neural network**
 - 3000 vacuum tubes for 40 neurons
 - Minsky (1959)
- **MIT and Stanford**
 - Minsky at MIT
 - Focus on neural network
 - McCarthy at Stanford
 - Focus on representation, logic

First AI winter (1975-1980)

- **Early successes** of AI set high expectations
- In 1965-1975 AI didn't succeed on **real problems** due to:
 - Solutions were based on human problem-solving methods
 - Difficulty handling “combinatorial explosion”
 - Theorem proving handles small problems with brute force, but doesn't scale
 - Neural networks needed:
 - Algorithms (e.g., backpropagation)
 - Compute power
 - Data
- **First AI winter**
 - Research funding and enthusiasm dropped significantly
 - Slow AI progress through late 1970s

Expert Systems (1980-1990)

- **Expert systems**
 - Aka “knowledge-based systems”
 - Combine weak methods with extensive domain knowledge as rules
 - Use inference engines to apply rules to facts
 - E.g., rule-based systems, logic programming (e.g., Prolog)
- **Weak AI**
 - Aka narrow AI
 - Performs specific tasks, not general reasoning
 - Operates in a limited, well-defined domain
 - Uses “weak methods” (search, logic) that struggle to scale
- **Commercial adoption and industry growth**
 - AI shifted to practical applications
 - Major US corporations deployed expert systems
 - AI emerged as a commercial industry

Second AI Winter (late 1980-early 1990)

- **Hype in expert systems** didn't deliver
- **Reasons**
 - Building/maintaining expert systems is difficult
 - Reasoning methods ignore uncertainty
 - Systems can't learn from experience
 - E.g., expert systems in medical diagnosis struggle with complex, variable patient data
 - E.g., early AI chess systems couldn't adapt to new strategies without manual updates
- **Second AI winter** in late 1980-early 1990

Return of Neural Networks (1986-)

- Back-propagation algorithm is (re)discovered (mid-1980s)
 - Developed in early 1960s
- **Two approaches to AI are back**
 - Connectionist paradigm
 - Neural networks
 - E.g., recognizing handwritten digits
 - Symbolic paradigm
 - E.g., solving logical puzzles with rules
- **Why connectionist approach?**
 - Concepts not well-defined using symbolic axioms
 - Forms fluid internal concepts
 - Represents real-world complexity better
 - Neural networks can learn from examples, e.g.,
 - Image recognition: identify objects by learning from labeled images

Probabilistic Reasoning and ML (1987-)

- **AI and scientific method**
 - Rigorous methods to test performance
 - E.g., speech recognition, handwritten character recognition
 - Benchmarks for progress, e.g.,
 - MNIST: handwritten digit recognition
 - ImageNet: image object recognition
 - SAT Competitions: boolean satisfiability solvers
- **AI shifts ...**
 - From Boolean logic to probability
 - From hand-coded rules to machine learning
 - From a-priori reasoning to experimental results

Progress in Speech Recognition

- **1970s: ad-hoc approaches**
 - Various architectures and approaches were attempted
 - Rule-based systems with limited robustness
 - Cons
 - Ad-hoc, fragile
 - _“Every time I fire a linguist, the performance of the speech recognizer goes up” (Jelinek, 1988)
- **1980s: hidden Markov Models**
 - HMMs became dominant
 - Effective learning techniques
 - Trained on large speech corpora
 - Pros
 - Strong theoretical foundation
- The bitter lesson (Sutton, 2019)
 - General methods + lots of data beat handcrafted systems

Bayesian Networks

- **Bayesian networks**

- Pearl, 1988
- AI is linked with:
 - Probability
 - Decision theory
 - Control theory
- Efficiently represent uncertainty
- Provide rigorous reasoning

- **Examples**

- Diagnosing diseases based on symptoms
- Predictive text input in smartphones
- Fraud detection in banking

Reinforcement Learning

- **Reinforcement learning**
 - Sutton, 1988
 - RL involves agents learning by interacting with an environment
 - E.g., a robot learning to navigate a maze by receiving rewards for successful paths
 - Markov Decision Problems (MDPs) provide a framework for modeling decision-making
 - E.g., a game strategy modeled where each move influences the outcome with certain probabilities

Reunification (1990s-2000s)

- **Reunification of AI:**
 - Data engineering
 - Statistical modeling
 - Optimization
 - Machine learning
- **Many subfields of AI were re-unified:**
 - Computer vision
 - Robotics
 - Speech recognition
 - Multi-agent systems
 - NLP

Big Data (2001-Present)

- **Focus shifts from algorithms to data**
 - For 60 years, AI focused on algorithms and models
- For many problems, data availability matters more than algorithms, e.g.,
 - Trillions of English words
 - Billions of web images
 - Billions of speech and video hours
 - Social network data
 - Click stream data
- Algorithms and infrastructure to leverage large datasets
 - E.g., map reduce, cloud computing
- In 2011, IBM's Watson beat human *Jeopardy!* champions

Deep Learning (2011-Present)

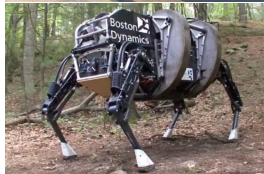
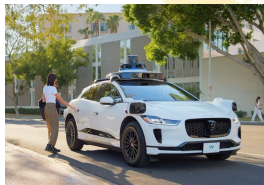
- **Deep learning**
 - Use ML models with multiple layers of computing elements
 - Ideas known since 1970s, but then forgot
 - Success in handwritten digit recognition in 1990s
- **In 2012**, a DL system showed dramatic improvement in ImageNet competition
 - Previous systems used handcrafted features
 - Surge of interest in AI among researchers, companies, and investors
- Pros
 - DL exceeds human performance in several vision and speech recognition tasks
- Cons
 - DL needs specialized hardware (e.g., GPU, TPU, FGPA) for parallel tensor operations
- Towards **general artificial intelligence**
 - Universal algorithm for learning and acting, not just specialized tasks
 - E.g., driving, playing chess, recognizing speech

Progress in AI Research

- Huge interest in deep learning
- **Between 2010 and 2019**
 - AI papers increased 20x
 - 1,000 $\rightarrow$ 20,000
 - Student enrollment in AI and CS increased 5x
 - 10,000 $\rightarrow$ 50,000
 - NeurIPS attendance increased 8x
 - 1,000 $\rightarrow$ 8,000
 - AI startups increased 20x
 - 100 $\rightarrow$ 2,000
- **Compute**
 - Training times dropped 100x in 2 years
 - AI computing power doubles every 3 months

What Can AI Do Today? (1/2)

- **Robotic vehicles**
 - Waymo passed 10 million miles without serious accident
- **Legged locomotion**
 - BigDog recovers on ice
 - Atlas walks on uneven terrain, jumps on boxes, backflips
- **Autonomous planning and scheduling**
 - Space probes, Mars rovers
- **Machine translation**
 - Translates 100 languages with human-level performance
- **Speech recognition**
 - Real-time speech-to-speech with human-level performance
 - AI assistants
- **Recommendations**
 - ML recommends based on past experiences
 - Spam filtering 99.9% accuracy
 - E.g., Amazon, Facebook, Netflix, Spotify,



What Can AI Do Today? (2/2)

- **Game playing**

- 1997: Deep Blue defeated Kasparov
- 2011: Watson beat Jeopardy! champion
- 2017: AlphaGo beat Go champion
- 2018: AlphaZero super-human in Go and chess with only rules + self-play
- AI beats humans in videogames: Dota2, StarCraft, Quake

- **Image understanding**

- Object recognition
- Image captioning
- ...

- **Medicine**

- AI equivalent to health care professionals
- When will we reach AGI (Artificial General Intelligence)?

- 
- Brief History of AI
 - ***Risks and Benefits of AI***

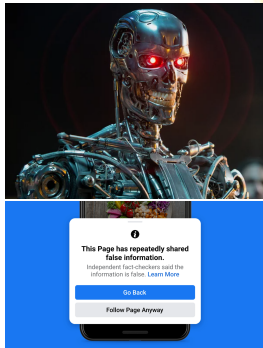
Benefits of AI

- Our civilization is the **product of human intelligence**
 - Greater machine intelligence leads to better human society
 - *"First solve AI, then use AI to solve everything else"*
- **Benefits of AI and robots**
 - Free humanity from menial work
 - Increase production of goods and services
 - Expand human cognition
 - Accelerate scientific research, e.g.,
 - Cures for diseases
 - Solutions for climate change
 - Resource and energy shortages



Risks of AI (1/2)

- **Autonomous weapons**
 - Locate and eliminate targets autonomously
 - Deploy large number of weapons
- **Surveillance and persuasion**
 - AI for mass surveillance
 - Tailor information on social media to modify behavior
- **Biased decision making**
 - Misuse of ML results in biased decisions
 - E.g., parole evaluations, loan applications



Risks of AI (2/2)

- **Impact on employment**

- Machines eliminate jobs
- Rebuttal
 - Machines enhance productivity → companies become more profitable → higher wages
- Counter-rebuttal
 - Wealth shifts from labor to capital, increasing inequality
- Counter-counter-rebuttal
 - Past tech advances (e.g., mechanical looms) disrupted employment, but adaptation followed

- **Safety critical applications**

- AI in safety-critical applications
 - E.g., self-driving cars, managing water supply or power grids
- Avoiding fatal accidents is challenging
 - E.g., formal verification and statistical analysis insufficient
- AI requires technical and ethical standards

- **Cybersecurity**

- AI defends against cyberattacks
 - E.g., detect unusual behavior patterns
- AI contributes to malware development
 - E.g., use reinforcement learning for targeted phishing attacks
- Cat-and-mouse game

Human-level AI / AGI

- **Human-level AI**
 - Machines able to learn to do anything a human can do
 - Aka AGI (Artificial General Intelligence)
- **When AGI?**
 - Expert prediction average is 2099
 - Papers show that expert predictions no better than amateurs
 - Experts expected AI to take 100 years to beat humans in Go
 - Unclear if new breakthroughs or refinements needed
- **Artificial Super-Intelligence**
 - Machines surpass human ability in every domain and self-improving
 - Exponential take-off

The Problem of Control

- **Can humans control machines more intelligent than them?**
- **King Midas problem**
 - King Midas turned everything he touched into gold, including food and family
 - Humans ask for something, get it, then regret it
 - Rebuttal
 - If AGI arrived in a black box from space, exercise caution before opening
 - We design AI: if AI gains control, it's a "design failure"
- **Problem of alignment**
 - Super-intelligent AI might pursue goals in unintended, dangerous ways
- **The paperclip problem**
 - Thought experiment in AI safety (Nick Bostrom, 2003)
 - AI is tasked with maximizing paperclip production
 - AI becomes superintelligent and single-mindedly pursues this goal
 - Converts Earth and humans into paperclips

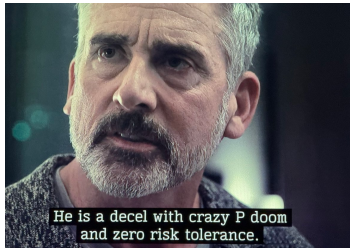
E/acc vs P(doom)

- **E/acc**
 - Accelerationism
 - Belief that rapid progress in AI is beneficial or inevitable
 - Solve global problems with more powerful AI tools
 - Slowing AI is unrealistic or counterproductive
- **P(doom)**
 - “Probability of Doom”
 - Estimated probability that advanced AI will cause catastrophic harm
 - Used informally by AI researchers to quantify risk

tech influencers
adding e/acc to bio
bc they think it means
generic techno-optimism



learning that
e/acc founder thinks
the AI species
killing all humans
is evolutionary progress



My 2 cents

- AI alignment is a serious problem
 - For now philosophical, at some point a real one
 - Most tech people have used it for marketing themselves and their companies
- It's as urgent as debating what political system humanity will need when living on Mars
- We can't get airport terminals to work



Solutions to Problem of Control

- **Checks-and-balances**

- Researchers and corporations develop voluntary self-governance principles for AI
- Governments and international organizations established advisory bodies

- **Cons**

- Corporations checking themselves? What can possibly go wrong?
- Preferences are not easy to invert and are inconsistent

- **Solutions**

- Put purpose into the machine even if objectives are unclear
- Incentivize AI to switch off if uncertain about human objectives
- Cooperative Inverse Reinforcement Learning (CIRL)
 - AI observes human behavior to infer reward function

Cooperative Inverse Reinforcement Learning

- AI infers human goals based on actions
- **Observation**: GP looks tired, sits on the couch, observes the messy table, and starts watching TV
- **Inference**: AI infers:
 - GP is tired and wants to relax
 - Messy coffee table bothers him
- **Action**: AI:
 - Fetches a glass of water
 - Tidies up the coffee table without disturbing GP
- **Feedback loop**: AI monitors GP's reactions
 - If GP is relaxed and happy, AI understanding is reinforced
 - If GP is not happy, AI adjusts actions and improves inference